THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 8	$F = 2 + E - V = 8$
2	**(d)** 16	Even number of values: mean of the two middle values
3	**(a)** $5n$	$5(n+3) - 15 = 5n$
4	**(b)** 15	$b = A/l = 360/24$
5	**(a)** 4	Each segment becomes 4: $4^1 = 4$
6	**(a)** 20	Total − known sum
7	**(b)** $3n + 3$	$n + (n+1) + \cdots = 3n + 3$
8	**(a)** 9	$h = 2A/b = 2 \times 72/16$
9	**(d)** $\frac{9}{16}$	Each stage keeps $\frac{3}{4}$: $(3/4)^2$
10	**(d)** 22	New sum 132, new count 6
11	**(a)** 20	Top $= a + 3b + 3c + d$
12	**(c)** 100 cm²	$\frac{1}{2}d(h_1 + h_2) = \frac{1}{2} \times 20 \times (6+4)$
13	**(d)** 6	$6(n-2)^2 = 6 \times 1$
14	**(b)** Between week 2 and week 3	Growth in each week: 4, 7, 4, 3, 6; largest is 7
15	**(d)** 90	Sum $= 9\times$ middle date

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Edges $= 3n = 21$ so $n = 7$ (1); faces $= n + 2 = 9$ (1).	1+1
17	Ascending: 12, 21, 34, 38, 44, 45 (1); mean of the two middle values $= 36$ (1).	1+1
18	Sum $= 16 + 61 = 77$ (1) $= 11 \times 7$ (1).	1+1
19	$b = A/l = 1200 \div 48 = 25$ m (1); perimeter $= 2(48 + 25) = 146$ m (1).	1+1
20	$F = 10$, $V = 10$, $E = 18$ (1); $F + V - E = 10 + 10 - 18 = 2$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	(i) $12(n-2) = 12 \times 1 = 12$ (1+1); (ii) $6(n-2)^2 = 6 \times 1 = 6$ (1).	2+1
22	Mean $= 61$ (1); median $= 66.5$ (1); range $= 58$ (1).	1+1+1
23	Row 2: 5, 4, 5 (1); row 3: 9, 9 (1); top $= 18$ (1).	1+1+1
24	Area $= 32 \times 18 = 576$ m² (1); cost $= 576 \times 12$ (1) $=$ ₹ 6912 (1).	1+1+1
	OR: Segments $= 4^3 = 64$ (1); each $= L/3^3 = 1$ cm (1); total $= 64$ cm (1).	1+1+1
25	$\sum xf = 51$ (1); $\sum f = 10$ (1); mean $= 5.1$ (1).	1+1+1
	OR: $83 - 38 = 45$ (2) $= 9 \times 5$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Prism: $F = 5$, $V = 6$, $E = 9$ (2); $F + V - E = 5 + 6 - 9 = 2$ (1); pyramid: $F = 4$, $V = 4$, $E = 6$ (1); $F + V - E = 4 + 4 - 6 = 2$ (1).	2+1+1+1
	OR: (i) mean $= 130$ (1); (ii) rises -98, -39, 129, -25, 3; greatest in month September (2); (iii) range $= 137$ (2).	1+2+2
27	Terms $n, n+1, \ldots, n+2$ (1); sum $3n + 3$ (2); for $n = 14$ the numbers add to 45 and $3 \times 14 + 3 = 45$ (2).	1+2+2
	OR: Trapezium $= \frac{1}{2}(20 + 12) \times 9 = 144$ cm² (2); parallelogram $= 12 \times 9 = 108$ cm² (1); left over $= 36$ cm² (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 9720	$18 \times 12 \times 45$
2	**(c)** two circles and one rectangle	Unrolling the curved surface gives a rectangle
3	**(b)** 7	The value with most dots is 7
4	**(b)** 1	$n^2 - (n-1)(n+1) = 1$
5	**(a)** 20000	$1\ \text{m}^2 = 10000\ \text{cm}^2$
6	**(b)** circle (with a dot at the centre)	Looking down, the base circle with the apex at centre
7	**(c)** 6	$\sum xf = 72, \sum f = 12$
8	**(b)** $4n$	Expand: $4an = 4 \times 1n$
9	**(c)** 8	$a + b = 2A/h = 20$
10	**(c)** 7	$F = 2 + E - V = 7$
11	**(c)** 19	Even number of values: mean of the two middle values
12	**(a)** $4n$	$4(n+6) - 24 = 4n$
13	**(a)** 11	$b = A/l = 352/32$
14	**(c)** 16	Each segment becomes 4: $4^2 = 16$
15	**(b)** 26	Total − known sum

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Total $= 4 \times 27 = 108$ (1); fourth $= 108 - 76 = 32$ (1).	1+1
17	$3(n+4) - 12 = 3n$ (1); $\div 3$ gives n (1).	1+1
18	Area $= 250 \times 80 = 20000$ cm² (1) $= 2$ m² (1).	1+1
19	$V = 2 + E - F$ (1) $= 14$ (1).	1+1
20	Old mean $= 24$ (1); new mean $= 28$ (adds 4) (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$A = \frac{1}{2}(a+b)h$ (1); $130 = \frac{1}{2} \times 26 \times h$ (1); $h = 10$ cm (1).	1+1+1
22	Shaded area $= 64 \times (3/4)^2 = 36$ cm² (2); removed $= 28$ cm² (1).	2+1
23	Sum $= 120$ (1); new sum $= 108$ over 4 values (1); mean $= 27$ (1).	1+1+1
24	Dates: 3, 4, 5, 10, 11, 12, 17, 18, 19 (1); sum $= 99$ (1); $= 9 \times 11$ (1).	1+1+1
	OR: Area $= \frac{1}{2} \times 12 \times 16 = 96$ cm² (2); side $= \sqrt{6^2 + 8^2} = 10$ cm (1).	2+1
25	(i) $12(n-2) = 12 \times 2 = 24$ (1+1); (ii) $6(n-2)^2 = 6 \times 4 = 24$ (1).	2+1
	OR: Mean $= 48$ (1); median $= 36$ (1); range $= 74$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Three faces: 8 (1); two faces: $12(9-2) = 84$ (1); one face: $6(9-2)^2 = 294$ (1); none: $(9-2)^3 = 343$ (1); total $= 8 + 84 + 294 + 343 = 729$ (1).	1+1+1+1+1
	OR: Mean A $= 41$, mean B $= 37$ (2); range A $= 62$, range B $= 29$ (2); more consistent: B (1).	2+2+1
27	$(2n+10)/2 - n = n + 5 - n = 5$ (3); for $n = 7$: $(2 \times 7 + 10)/2 - 7 = 5$ (2).	3+2
	OR: Area $= 12 \times 9 = 108$ m² (1); length of carpet $= \text{area}/\text{width} = 54$ m (2); cost $= 54 \times 80 =$ ₹ 4320 (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** $4n+6$	$n+(n+1)+\cdots=4n+6$
2	**(d)** 13	$h=2A/b=2\times130/20$
3	**(c)** $\frac{27}{64}$	Each stage keeps $\frac{3}{4}$: $(3/4)^3$
4	**(d)** 30	New sum 180, new count 6
5	**(d)** 23	Top $=a+3b+3c+d$
6	**(a)** 144 cm²	$\frac{1}{2}d(h_1+h_2)=\frac{1}{2}\times24\times(5+7)$
7	**(d)** 24	$6(n-2)^2=6\times4$
8	**(a)** Between week 3 and week 4	Growth in each week: 3, 2, 6, 0, 0; largest is 6
9	**(b)** 108	Sum $=9\times$ middle date
10	**(b)** 15000	$25\times10\times60$
11	**(a)** one circle and one sector	Curved surface opens into a sector
12	**(d)** 3	The value with most dots is 3
13	**(a)** 1	$n^2-(n-1)(n+1)=1$
14	**(d)** 30000	$1\ \text{m}^2=10000\ \text{cm}^2$
15	**(d)** rectangle	Side view of upright cylinder

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Numbers $2n+1$ and $2n+3$ (1); sum $=4n+4=4(n+1)$ (1).	1+1
17	Area $=\frac{1}{2}d(h_1+h_2)$ (1) $=\frac{1}{2}\times26\times13=169$ cm² (1).	1+1
18	Edges $=3n=27$ so $n=9$ (1); faces $=n+2=11$ (1).	1+1
19	Ascending: 9, 10, 19, 35, 42, 60 (1); mean of the two middle values $=27$ (1).	1+1
20	Sum $=38+83=121$ (1) $=11\times11$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Row 2: 7, 5, 4 (1); row 3: 12, 9 (1); top $=21$ (1).	1+1+1
22	Area $=45\times20=900$ m² (1); cost $=900\times8$ (1) $=$ ₹ 7200 (1).	1+1+1
23	Segments $=4^4=256$ (1); each $=L/3^4=1$ cm (1); total $=256$ cm (1).	1+1+1
24	$\sum xf=240$ (1); $\sum f=10$ (1); mean $=24$ (1).	1+1+1
	OR: $94-49=45$ (2) $=9\times5$ (1).	2+1
25	$A=\frac{1}{2}(a+b)h$ (1); $144=\frac{1}{2}\times36\times h$ (1); $h=8$ cm (1).	1+1+1
	OR: Shaded area $=256\times(3/4)^2=144$ cm² (2); removed $=112$ cm² (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Prism: $F=6$, $V=8$, $E=12$ (2); $F+V-E=6+8-12=2$ (1); pyramid: $F=5$, $V=5$, $E=8$ (1); $F+V-E=5+5-8=2$ (1).	2+1+1+1
	OR: (i) mean $=121$ (1); (ii) rises -100, -9, 90, 27, 20; greatest in month September (2); (iii) range $=137$ (2).	1+2+2
27	Terms $n, n+1, \ldots, n+3$ (1); sum $4n+6$ (2); for $n=9$ the numbers add to 42 and $4\times9+6=42$ (2).	1+2+2
	OR: Trapezium $=\frac{1}{2}(26+10)\times8=144$ cm² (2); parallelogram $=10\times8=80$ cm² (1); left over $=64$ cm² (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND